

ANNEX I

SUMMARY OF PRODUCT CHARACTERISTICS

▼ This medicinal product is subject to additional monitoring. This will allow quick identification of new safety information. Healthcare professionals are asked to report any suspected adverse reactions. See section 4.8 for how to report adverse reactions.

1. NAME OF THE MEDICINAL PRODUCT

Qdenga powder and solvent for solution for injection

Qdenga powder and solvent for solution for injection in pre-filled syringe

Dengue tetravalent vaccine (live, attenuated)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

After reconstitution, 1 dose (0.5 mL) contains:

Dengue virus serotype 1 (live, attenuated)*: $\geq 3.3 \log_{10}$ PFU**/dose

Dengue virus serotype 2 (live, attenuated)#: $\geq 2.7 \log_{10}$ PFU**/dose

Dengue virus serotype 3 (live, attenuated)*: $\geq 4.0 \log_{10}$ PFU**/dose

Dengue virus serotype 4 (live, attenuated)*: $\geq 4.5 \log_{10}$ PFU**/dose

*Produced in Vero cells by recombinant DNA technology. Genes of serotype-specific surface proteins engineered into dengue type 2 backbone. This product contains genetically modified organisms (GMOs).

#Produced in Vero cells by recombinant DNA technology

**PFU = Plaque-forming units

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Powder and solvent for solution for injection.

Prior to reconstitution, the vaccine is a white to off-white coloured freeze-dried powder (compact cake).

The solvent is a clear, colourless solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Qdenga is indicated for the prevention of dengue disease in individuals from 4 years of age.

The use of Qdenga should be in accordance with official recommendations.

4.2 Posology and method of administration

Posology

Individuals from 4 years of age

Qdenga should be administered as a 0.5 mL dose at a two-dose (0 and 3 months) schedule.

The need for a booster dose has not been established.

Other paediatric population (children <4 years of age)

The safety and efficacy of Qdenga in children aged less than 4 years has not yet been established. Currently available data are described in section 4.8 but no recommendation on a posology can be made.

Elderly

No dose adjustment is required in elderly individuals ≥ 60 years of age. See section 4.4.

Method of administration

After complete reconstitution of the lyophilised vaccine with the solvent, Qdenga should be administered by subcutaneous injection preferably in the upper arm in the region of deltoid.

Qdenga must not be injected intravascularly, intradermally or intramuscularly.

The vaccine should not be mixed in the same syringe with any other vaccines or other parenteral medicinal products.

For instructions on reconstitution of Qdenga before administration, see section 6.6.

4.3 Contraindications

- Hypersensitivity to the active substances or to any of the excipients listed in section 6.1 or hypersensitivity to a previous dose of Qdenga.
- Individuals with congenital or acquired immune deficiency, including those receiving immunosuppressive therapies such as high doses of systemic corticosteroids (e.g. 20 mg/day or 2 mg/kg body weight/day of prednisone for 2 weeks or more) within 4 weeks prior to vaccination, or any other medicinal product with known immunosuppressive properties including chemotherapy. The time to avoid vaccination after immunosuppressive treatment should be considered on an individual basis.
- Individuals with symptomatic HIV infection or with asymptomatic HIV infection when accompanied by evidence of impaired immune function.
- Pregnant women (see section 4.6).
- Breast-feeding women (see section 4.6).

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

General recommendations

Anaphylaxis

Anaphylaxis has been reported in individuals who have received Qdenga. The majority of cases occurred within 30 minutes following vaccination. As with all injectable vaccines, appropriate medical treatment and supervision must always be readily available in the event of a rare anaphylactic reaction following administration of the vaccine.

Review of medical history

Vaccination should be preceded by a review of the individual's medical history (especially with regard to previous vaccination and possible hypersensitivity reactions which occurred after vaccination).

Concurrent illness

Vaccination with Qdenga should be postponed in subjects suffering from an acute severe febrile illness. The presence of a minor infection, such as a cold, should not result in a deferral of vaccination.

Limitations of vaccine effectiveness

A protective immune response with Qdenga may not be elicited in all vaccinees against all serotypes of dengue virus and may decline over time (see section 5.1). It is currently unknown whether a lack of protection could result in an increased severity of dengue. It is recommended to continue personal protection measures against mosquito bites after vaccination. Individuals should seek medical care if they develop dengue symptoms or dengue warning signs.

There are no data on the use of Qdenga in subjects above 60 years of age and limited data in patients with chronic medical conditions.

Anxiety-related reactions

Anxiety-related reactions, including vasovagal reactions (syncope), hyperventilation or stress-related reactions may occur in association with vaccination as a psychogenic response to the needle injection. It is important that precautions are in place to avoid injury from fainting.

Women of childbearing potential

As with other live attenuated vaccines, women of childbearing potential should avoid pregnancy for at least one month following vaccination (see sections 4.6 and 4.3).

Other

Qdenga must not be administered by intravascular, intradermal or intramuscular injection.

Excipients

Qdenga contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially 'sodium-free'.

Qdenga contains less than 1 mmol potassium (39 mg) per dose, that is to say essentially 'potassium-free'.

4.5 Interaction with other medicinal products and other forms of interaction

For patients receiving treatment with immunoglobulins or blood products containing immunoglobulins, such as blood or plasma, it is recommended to wait for at least 6 weeks, and

preferably for 3 months, following the end of treatment before administering Qdenga, in order to avoid neutralisation of the attenuated viruses contained in the vaccine.

Qdenga should not be administered to subjects receiving immunosuppressive therapies such as high doses of systemic corticosteroids within 4 weeks prior to vaccination, or any other medicinal product with known immunosuppressive properties including chemotherapy (see section 4.3). The time to avoid vaccination after immunosuppressive treatment should be considered on an individual basis.

Use with other vaccines

If Qdenga is to be given at the same time as another injectable vaccine, the vaccines should always be administered at different injection sites.

Qdenga may be administered concomitantly with a hepatitis A vaccine. Coadministration has been studied in adults.

Qdenga may be administered concomitantly with a yellow fever vaccine. In a clinical study involving approximately 300 adult subjects who received Qdenga concomitantly with yellow fever 17D vaccine, there was no effect on yellow fever seroprotection rate. Dengue antibody responses were decreased following concomitant administration of Qdenga and yellow fever 17D vaccine. The clinical significance of this finding is unknown.

Qdenga may be administered concomitantly with a human papillomavirus (HPV) vaccine (see section 5.1).

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Women of childbearing potential should avoid pregnancy for at least one month following vaccination. Women who intend to become pregnant should be advised to delay vaccination (see sections 4.4 and 4.3).

Pregnancy

Animal studies are insufficient with respect to reproductive toxicity (see section 5.3).

There is limited amount of data from the use of Qdenga in pregnant women. These data are not sufficient to conclude on the absence of potential effects of Qdenga on pregnancy, embryo-foetal development, parturition and post-natal development.

Qdenga is a live attenuated vaccine, therefore Qdenga is contraindicated during pregnancy (see section 4.3).

Breast-feeding

It is unknown whether Qdenga is excreted in human milk. A risk to the newborns/infants cannot be excluded.

Qdenga is contraindicated during breast-feeding (see section 4.3).

Fertility

Animal studies are insufficient with respect to reproductive toxicity (see section 5.3).

No specific studies have been performed on fertility in humans.

4.7 Effects on ability to drive and use machines

Qdenga has minor influence on the ability to drive and use machines.

4.8 Undesirable effects

Summary of the safety profile

In clinical studies, the most frequently reported reactions in subjects 4 to 60 years of age were injection site pain (50%), headache (35%), myalgia (31%), injection site erythema (27%), malaise (24%), asthenia (20%) and fever (11%).

These adverse reactions usually occurred within 2 days after the injection, were mild to moderate in severity, had a short duration (1 to 3 days) and were less frequent after the second injection of Qdenga than after the first injection.

Vaccine viraemia

In clinical study DEN-205, transient vaccine viraemia was observed after vaccination with Qdenga in 49% of study participants who had not been infected with dengue before and in 16% of study participants who had been infected with dengue before. Vaccine viraemia usually started in the second week after the first injection and had a mean duration of 4 days. Vaccine viraemia was associated with transient, mild to moderate symptoms, such as headache, arthralgia, myalgia and rash in some subjects that may also occur with dengue. Vaccine viraemia was rarely detected after the second dose. Dengue diagnostic tests may be positive during vaccine viraemia and cannot be used to distinguish vaccine viraemia from wild type dengue infection.

Tabulated list of adverse reactions

Adverse reactions associated with Qdenga obtained from clinical studies and post-authorisation experience are tabulated below (**Table 1**).

The safety profile presented below is based on data generated in placebo-controlled clinical studies and post-authorisation experience. Pooled analysis of clinical studies included data from 14,627 study participants aged 4 to 60 years (13,839 children and 788 adults) who have been vaccinated with Qdenga. This included a reactogenicity subset of 3,830 participants (3,042 children and 788 adults).

Adverse reactions are listed according to the following frequency categories:

Very common: $\geq 1/10$

Common: $\geq 1/100$ to $< 1/10$

Uncommon: $\geq 1/1,000$ to $< 1/100$

Rare: $\geq 1/10,000$ to $< 1/1,000$

Very rare: $< 1/10,000$

Not known: cannot be estimated from the available data

Table 1: Adverse reactions from clinical studies (age 4 to 60 years) and post-authorisation experience (age 4 years and older)

MedDRA System Organ Class	Frequency	Adverse Reactions
Infections and infestations	Very common	Upper respiratory tract infection ^a
	Common	Nasopharyngitis Pharyngotonsillitis ^b
	Uncommon	Bronchitis Rhinitis
Blood and lymphatic system disorders	Very rare	Thrombocytopenia ^c

MedDRA System Organ Class	Frequency	Adverse Reactions
Immune system disorders	Not known	Anaphylactic reaction, including anaphylactic shock ^c
Metabolism and nutrition disorders	Very common	Decreased appetite ^d
Psychiatric disorders	Very common	Irritability ^d
Nervous system disorders	Very common	Headache Somnolence ^d
	Uncommon	Dizziness
Eye disorders	Not known	Eye pain ^c
Gastrointestinal disorders	Uncommon	Diarrhoea Nausea Abdominal pain Vomiting
Skin and subcutaneous tissue disorders	Uncommon	Rash ^e Pruritus ^f Urticaria
	Rare	Petechiae ^c
	Very rare	Angioedema
Musculoskeletal and connective tissue disorders	Very common	Myalgia
	Common	Arthralgia
General disorders and administration site conditions	Very common	Injection site pain Injection site erythema Malaise Asthenia Fever
	Common	Injection site swelling Injection site bruising ^f Injection site pruritus ^f Influenza like illness
	Uncommon	Injection site haemorrhage ^f Fatigue ^f Injection site discolouration ^f

^a Includes upper respiratory tract infection and viral upper respiratory tract infection

^b Includes pharyngotonsillitis and tonsillitis

^c Adverse reaction observed post-authorisation

^d Collected in children below 6 years of age in clinical studies

^e Includes rash, viral rash, rash maculopapular, rash pruritic

^f Reported in adults in clinical studies

Paediatric population

Paediatric data in subjects 4 to 17 years of age

Pooled safety data from clinical trials are available for 13839 children (9210 aged 4 to 11 years and 4629 aged 12 to 17 years). This includes reactogenicity data collected in 3042 children (1865 aged 4 to 11 years and 1177 aged 12 to 17 years).

Frequency, type and severity of adverse reactions in children were largely consistent with those in adults. Adverse reactions reported more commonly in children than in adults were fever (11% versus 3%), upper respiratory tract infection (11% versus 3%), nasopharyngitis (6% versus 0.6%), pharyngotonsillitis (2% versus 0.3%), and influenza like illness (1% versus 0.1%). Adverse reactions reported less commonly in children than adults were injection site erythema (2% versus 27%), nausea (0.03% versus 0.8%) and arthralgia (0.03% versus 1%).

The following reactions were collected in 357 children below 6 years of age vaccinated with Qdenga:

decreased appetite (17%), somnolence (13%) and irritability (12%).

Paediatric data in subjects below 4 years of age, i.e. outside the age indication

Reactogenicity in subjects below 4 years of age was assessed in 78 subjects who received at least one dose of Qdenga of which 13 subjects received the indicated 2-dose regimen. Reactions reported with very common frequency were irritability (25%), fever (17%), injection site pain (17%) and loss of appetite (15%). Somnolence (8%) and injection site erythema (3%) were reported with common frequency. Injection site swelling was not observed in subjects below 4 years of age.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system listed in [Appendix V](#).

4.9 Overdose

No cases of overdose have been reported.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Vaccines, Viral vaccines, ATC code: J07BX04

Mechanism of action

Qdenga contains live attenuated dengue viruses. The primary mechanism of action of Qdenga is to replicate locally and elicit humoral and cellular immune responses against the four dengue virus serotypes.

Clinical efficacy

The clinical efficacy of Qdenga was assessed in study DEN-301, a pivotal Phase 3, double-blind, randomised, placebo-controlled study conducted across 5 countries in Latin America (Brazil, Colombia, Dominican Republic, Nicaragua, Panama) and 3 countries in Asia (Sri Lanka, Thailand, the Philippines). A total of 20,099 children aged between 4 and 16 years were randomised (2:1 ratio) to receive Qdenga or placebo, regardless of previous dengue infection.

Efficacy was assessed using active surveillance across the entire study duration. Any subject with febrile illness (defined as fever $\geq 38^{\circ}\text{C}$ on any 2 of 3 consecutive days) was required to visit the study site for dengue fever evaluation by the investigator. Subjects/guardians were reminded of this requirement at least weekly to maximise the detection of all symptomatic virologically confirmed dengue (VCD) cases. Febrile episodes were confirmed by a validated, quantitative dengue RT-PCR to detect specific dengue serotypes.

Clinical efficacy data for subjects 4 to 16 years of age

The Vaccine Efficacy (VE) results, according to the primary endpoint (VCD fever occurring from 30 days to 12 months after the second vaccination) are shown in **Table 2**. The mean age of the per protocol trial population was 9.6 years (standard deviation of 3.5 years) with 12.7% subjects in the 4-5 years, 55.2% in the 6-11 years and 32.1% in the 12-16 years age-groups. Of these, 46.5% were in Asia and 53.5% were in Latin America, 49.5% were females and 50.5% were males. The dengue serostatus at baseline (before the first injection) was assessed in all subjects by microneutralisation test (MNT₅₀)

to allow Vaccine Efficacy (VE) assessment by baseline serostatus. The baseline dengue seronegativity rate for the overall per protocol population was 27.7%.

Table 2: Vaccine efficacy in preventing VCD fever caused by any serotype from 30 days to 12 months post second vaccination in study DEN-301 (Per Protocol Set)^a

	Qdenga N = 12,700 ^b	Placebo N = 6316 ^b
VCD fever, n (%)	61 (0.5)	149 (2.4)
Vaccine efficacy (95% CI) (%)	80.2 (73.3, 85.3)	
p-value	<0.001	

CI: confidence interval; n: number of subjects with fever; VCD: virologically confirmed dengue

^a The primary analysis of efficacy data were based on the Per Protocol Set, which consisted of all randomised subjects who did not have any major protocol violations, including not receiving both doses of the correct assignment of Qdenga or placebo

^b Number of subjects evaluated

VE results according to the secondary endpoints, preventing hospitalisation due to VCD fever, preventing VCD fever by serostatus, by serotype and preventing severe VCD fever are shown in **Table 3**. For severe VCD fever, two types of endpoints were considered: clinically severe VCD cases and VCD cases that met the 1997 WHO criteria for Dengue Haemorrhagic Fever (DHF). The criteria used in Trial DEN-301 for the assessment of VCD severity by an independent “Dengue Case severity Adjudication Committee” (DCAC) were based on the WHO 2009 guidelines. The DCAC assessed all cases of hospitalisation due to VCD utilizing predefined criteria which included an assessment of bleeding abnormality, plasma leakage, liver function, renal function, cardiac function, the central nervous system, and shock. In Trial DEN-301 VCD cases meeting the WHO 1997 criteria for DHF were identified using a programmed algorithm, i.e., without applying medical judgment. Broadly, the criteria included presence of fever lasting 2 to 7 days, haemorrhagic tendencies, thrombocytopenia, and evidence of plasma leakage.

Table 3: Vaccine efficacy in preventing hospitalisation due to VCD fever, VCD fever by dengue serotype, VCD fever by baseline dengue serostatus, and severe forms of dengue from 30 days to 18 months post second vaccination in study DEN-301 (Per Protocol Set)

	Qdenga N=12,700 ^a	Placebo N=6316 ^a	VE (95% CI)
VE in preventing hospitalisations due to VCD fever^b, n (%)			
Hospitalisations due to VCD fever ^c	13 (0.1)	66 (1.0)	90.4 (82.6, 94.7) ^d
VE in preventing VCD fever by dengue serotype, n (%)			
VCD fever caused by DENV-1	38 (0.3)	62 (1.0)	69.8 (54.8, 79.9)
VCD fever caused by DENV-2	8 (<0.1)	80 (1.3)	95.1 (89.9, 97.6)
VCD fever caused by DENV-3	63 (0.5)	60 (0.9)	48.9 (27.2, 64.1)
VCD fever caused by DENV-4	5 (<0.1)	5 (<0.1)	51.0 (-69.4, 85.8)
VE in preventing VCD fever by baseline dengue serostatus, n (%)			
VCD fever in all subjects	114 (0.9)	206 (3.3)	73.3 (66.5, 78.8)
VCD fever in baseline seropositive subjects	75 (0.8)	150 (3.3)	76.1 (68.5, 81.9)
VCD fever in baseline seronegative subjects	39 (1.1)	56 (3.2)	66.2 (49.1, 77.5)
VE in preventing DHF induced by any dengue serotype, n (%)			
Overall	2 (<0.1)	7 (0.1)	85.9 (31.9, 97.1)
VE in preventing severe dengue induced by any dengue serotype, n (%)			
Overall	2 (<0.1)	1 (<0.1)	2.3 (-977.5, 91.1)

VE: vaccine efficacy; CI: confidence interval; n: number of subjects; VCD: virologically confirmed dengue; DENV: dengue virus serotype

^a Number of subjects evaluated

^b key secondary endpoint

^c Most of the cases observed were due to DENV-2 (0 cases in Qdenga arm and 46 cases in Placebo arm)

^d p-value <0.001

Early onset of protection was seen with an exploratory VE of 81.1% (95% CI: 64.1%, 90.0%) against VCD fever caused by all serotypes combined from first vaccination until second vaccination.

Long term protection

In study DEN-301, a number of exploratory analyses were conducted to estimate long term protection from first dose up to 4.5 years after the second dose (**Table 4**).

Table 4: Vaccine efficacy in preventing VCD fever and hospitalisation overall, by baseline dengue serostatus, and against individual serotypes by baseline serostatus from first dose to 54 months post second dose in study DEN-301 (Safety Set)

	Qdenga n/N	Placebo n/N	VE (95% CI) in preventing VCD Fever ^a	Qdenga n/N	Placebo n/N	VE (95% CI) in preventing Hospitalisation due to VCD Fever ^a
Overall	442/13380	547/6687	61.2 (56.0, 65.8)	46/13380	142/6687	84.1 (77.8, 88.6)
Baseline Seronegative, N=5,546						
Any serotype	147/3714	153/1832	53.5 (41.6, 62.9)	17/3714	41/1832	79.3 (63.5, 88.2)
DENV-1	89/3714	79/1832	45.4 (26.1, 59.7)	6/3714	14/1832	78.4 (43.9, 91.7)
DENV-2	14/3714	58/1832	88.1 (78.6, 93.3)	0/3714	23/1832	100 (88.5, 100) ^b
DENV-3	36/3714	16/1832	-15.5 (-108.2, 35.9)	11/3714	3/1832	-87.9 (-573.4, 47.6)
DENV-4	12/3714	3/1832	-105.6 (-628.7, 42.0)	0/3714	1/1832	NP ^c
Baseline Seropositive, N=14,517						
Any serotype	295/9663	394/4854	64.2 (58.4, 69.2)	29/9663	101/4854	85.9 (78.7, 90.7)
DENV-1	133/9663	151/4854	56.1 (44.6, 65.2)	16/9663	24/4854	66.8 (37.4, 82.3)
DENV-2	54/9663	135/4854	80.4 (73.1, 85.7)	5/9663	59/4854	95.8 (89.6, 98.3)
DENV-3	96/9663	97/4854	52.3 (36.7, 64.0)	8/9663	15/4854	74.0 (38.6, 89.0)
DENV-4	12/9663	20/4854	70.6 (39.9, 85.6)	0/9663	3/4854	NP ^c

VE: vaccine efficacy, CI: confidence interval, VCD: virologically confirmed dengue, n: number of subjects, N: number of subjects evaluated, NP: not provided

^a Exploratory analyses; the study was neither powered nor designed to demonstrate a difference between the vaccine and the placebo group

^b Approximated using a one-sided 95% CI

^c VE estimate not provided since fewer than 6 cases, for both TDV and placebo, were observed

Additionally, VE in preventing DHF caused by any serotype was 70.0% (95% CI: 31.5%, 86.9%) and in preventing clinically severe VCD cases caused by any serotype was 70.2% (95% CI: -24.7%, 92.9%).

VE in preventing VCD was shown for all four serotypes in baseline dengue seropositive subjects. In baseline seronegative subjects, VE was shown for DENV-1 and DENV-2, but not suggested for DENV-3 and could not be shown for DENV-4 due to lower incidence of cases (**Table 4**).

A year-by-year analysis until four and a half years after the second dose was conducted (**Table 5**).

Table 5: Vaccine efficacy in preventing VCD fever and hospitalisation overall and by baseline dengue serostatus in yearly intervals 30 days post second dose in study DEN-301 (Per Protocol Set)

		VE (95% CI) in preventing VCD Fever N^a = 19,021	VE (95% CI) in preventing Hospitalisation due to VCD Fever N^a = 19,021
Year 1 ^b	Overall	80.2 (73.3, 85.3)	95.4 (88.4, 98.2)
	By baseline dengue serostatus		
	Seropositive	82.2 (74.5, 87.6)	94.4 (84.4, 98.0)
	Seronegative	74.9 (57.0, 85.4)	97.2 (79.1, 99.6)
Year 2 ^c	Overall	56.2 (42.3, 66.8)	76.2 (50.8, 88.4)
	By baseline dengue serostatus		
	Seropositive	60.3 (44.7, 71.5)	85.2 (59.6, 94.6)
	Seronegative	45.3 (9.9, 66.8)	51.4 (-50.7, 84.3)
Year 3 ^d	Overall	45.0 (32.9, 55.0)	70.8 (49.6, 83.0)
	By baseline dengue serostatus		
	Seropositive	48.7 (34.8, 59.6)	78.4 (57.1, 89.1)
	Seronegative	35.5 (7.4, 55.1)	45.0 (-42.6, 78.8)
Year 4 ^e	Overall	62.8 (41.4, 76.4)	96.4 (72.2, 99.5)
	By baseline dengue serostatus		
	Seropositive	64.1 (37.4, 79.4)	94.0 (52.2, 99.3)
	Seronegative	60.2 (11.1, 82.1)	NP ^f

VE: vaccine efficacy, CI: confidence interval, VCD: virologically confirmed dengue, NP: not provided, N: total number of subjects in the per analysis set, ^a number of subjects evaluated in each year is different.

^b Year 1 refers to 11 months starting 30 days after second dose.

^c Year 2 refers to 13 to 24 months after second dose.

^d Year 3 refers to 25 to 36 months after second dose.

^e Year 4 refers to 37 to 48 months after second dose.

^f VE estimate not provided since fewer than 6 cases, for both TDV and placebo, were observed.

Clinical efficacy for subjects from 17 years of age

No clinical efficacy study has been conducted in subjects from 17 years of age. The efficacy of Qdenga in subjects from 17 years of age is inferred from the clinical efficacy in 4 to 16 years of age by bridging of immunogenicity data (see below).

Immunogenicity

In the absence of correlates of protection for Dengue, the clinical relevance of immunogenicity data remains to be fully understood.

Immunogenicity data for subjects 4 to 16 years of age in endemic areas

The Geometric Mean Titres (GMTs) by baseline dengue serostatus in subjects 4 to 16 years of age in study DEN-301 are shown in **Table 6**.

Table 6: Immunogenicity by baseline dengue serostatus in study DEN-301 (Per Protocol Set for Immunogenicity)^a

	Baseline Seropositive		Baseline Seronegative	
	Pre-Vaccination N=1816*	1 month Post-Dose 2 N=1621	Pre-Vaccination N=702	1 month Post-Dose 2 N=641
DENV-1 GMT 95% CI	411.3 (366.0, 462.2)	2115.2 (1957.0, 2286.3)	5.0 NE**	184.2 (168.6, 201.3)
DENV-2 GMT 95% CI	753.1 (681.0, 832.8)	4897.4 (4645.8, 5162.5)	5.0 NE**	1729.9 (1613.7, 1854.6)
DENV-3 GMT 95% CI	357.7 (321.3, 398.3)	1761.0 (1645.9, 1884.1)	5.0 NE**	228.0 (211.6, 245.7)
DENV-4 GMT 95% CI	218.4 (198.1, 240.8)	1129.4 (1066.3, 1196.2)	5.0 NE**	143.9 (133.6, 155.1)

N: number of subjects evaluated; DENV: Dengue virus; GMT: Geometric Mean Titre; CI: confidence interval; NE: not estimated

^a The immunogenicity subset was a randomly selected subset of subjects, and the Per Protocol Set for Immunogenicity was the collection of subjects from that subset who also belong to the Per Protocol Set

* For DENV-2 and DENV-3: N= 1815

** All subjects had GMT values below LLOD (10), hence were reported as 5 with no CI values

Immunogenicity data for subjects 18 to 60 years of age in non-endemic areas

The immunogenicity of Qdenga in adults 18 to 60 years of age was assessed in DEN-304, a Phase 3 double-blind, randomized, placebo-controlled study in a non-endemic country (US). The post-dose 2 GMTs are shown in **Table 7**.

Table 7: GMTs of dengue neutralising antibodies in study DEN-304 (Per Protocol Set)

	Baseline Seropositive*		Baseline Seronegative*	
	Pre-Vaccination N=68	1 month Post-Dose 2 N=67	Pre-Vaccination N=379	1 month Post-Dose 2 N=367
DENV-1 GMT 95% CI	13.9 (9.5, 20.4)	365.1 (233.0, 572.1)	5.0 NE**	268.1 (226.3, 317.8)
DENV-2 GMT 95% CI	31.8 (22.5, 44.8)	3098.0 (2233.4, 4297.2)	5.0 NE**	2956.9 (2635.9, 3316.9)
DENV-3 GMT 95% CI	7.4 (5.7, 9.6)	185.7 (129.0, 267.1)	5.0 NE**	128.9 (112.4, 147.8)
DENV-4 GMT 95% CI	7.4 (5.5, 9.9)	229.6 (150.0, 351.3)	5.0 NE**	137.4 (121.9, 155.0)

N: number of subjects evaluated; DENV: Dengue virus; GMT: Geometric Mean Titre; CI: confidence interval; NE: not estimated

* Pooled data from Dengue tetravalent vaccine Lots 1, 2 and 3

** All subjects had GMT values below LLOD (10), hence were reported as 5 with no CI values

The bridging of efficacy is based on immunogenicity data and results from a non-inferiority analysis, comparing post-vaccination GMTs in the baseline dengue seronegative populations of DEN-301 and DEN-304 (**Table 8**). Protection against dengue disease is expected in adults although the actual magnitude of efficacy relative to that observed in children and adolescents is unknown.

Table 8: GMT ratios between baseline dengue seronegative subjects in studies DEN-301 (4-16 years) and DEN-304 (18-60 years) (Per Protocol Set for Immunogenicity)

GMT Ratio* (95% CI)	DENV-1	DENV-2	DENV-3	DENV-4
1m post-2 nd dose	0.69 (0.58, 0.82)	0.59 (0.52, 0.66)	1.77 (1.53, 2.04)	1.05 (0.92, 1.20)
6m post-2 nd dose	0.62 (0.51, 0.76)	0.66 (0.57, 0.76)	0.98 (0.84, 1.14)	1.01 (0.86, 1.18)

DENV: Dengue virus; GMT: Geometric Mean Titre; CI: confidence interval; m: month(s)

*Non-inferiority: upper bound of the 95% CI less than 2.0.

Long-term persistence of antibodies

The long-term persistence of neutralising antibodies was shown in study DEN-301, with titres remaining well above the pre-vaccination levels for all four serotypes, up to 51 months after the first dose.

Co-administration with HPV vaccine

In study DEN-308 involving approximately 300 subjects aged 9 to 14 years who received Qdenga concomitantly with a 9-valent HPV vaccine, there was no effect on the immune response to the HPV vaccine. The study only tested co-administration of the first doses of Qdenga and the 9-valent HPV vaccine. Non-inferiority of the Qdenga immune response, when Qdenga and the 9-valent HPV vaccine were co-administered, has not been directly assessed in the study. In the dengue seronegative study population, dengue antibody responses after co-administration were in the same range as those observed in the Phase 3 study (DEN-301) where efficacy against VCD and hospitalised VCD was shown.

5.2 Pharmacokinetic properties

No pharmacokinetic studies have been performed with Qdenga.

5.3 Preclinical safety data

Non-clinical safety data revealed no special hazard for humans based on conventional studies of single dose, local tolerance, repeated dose toxicity, and toxicity to reproduction and development. In a distribution and shedding study, there was no shedding of Qdenga RNA in faeces and urine, confirming a low risk for vaccine shedding to the environment or transmission from vaccinees. A neurovirulence study shows that Qdenga is not neurotoxic.

Although no relevant hazard was identified, the relevance of the reproductive toxicity studies is limited, since rabbits are not permissive for dengue virus infection.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Powder:

α,α-Trehalose dihydrate
 Poloxamer 407
 Human serum albumin
 Potassium dihydrogen phosphate
 Disodium hydrogen phosphate
 Potassium chloride
 Sodium chloride

Solvent:

Sodium chloride
 Water for injections

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other vaccine or medicinal products except for the solvent provided.

6.3 Shelf life

24 months.

After reconstitution with the solvent provided, Qdenga should be used immediately.

If not used immediately, Qdenga must be used within 2 hours.

Chemical and physical in-use stability have been demonstrated for 2 hours at room temperature (up to 32.5°C) from the time of reconstitution of the vaccine vial. After this time period, the vaccine must be discarded. Do not return it to the refrigerator.

From a microbiological point of view Qdenga should be used immediately. If not used immediately, in-use storage times and conditions are the responsibility of the user.

6.4 Special precautions for storage

Store in a refrigerator (2°C to 8°C). Do not freeze.

Store in the original package.

For storage conditions after reconstitution of Qdenga, see section 6.3.

6.5 Nature and contents of container

Qdenga powder and solvent for solution for injection:

- Powder (1 dose) in glass vial (Type-I glass), with a stopper (butyl rubber) and aluminium seal with green flip-off plastic cap + 0.5 mL solvent (1 dose) in glass vial (Type-I glass), with a stopper (bromobutyl rubber) and aluminium seal with purple flip-off plastic cap

Pack size of 1 or 10.

Qdenga powder and solvent for solution for injection in pre-filled syringe:

- Powder (1 dose) in vial (Type-I glass), with a stopper (butyl rubber) and aluminium seal with green flip-off plastic cap + 0.5 mL solvent (1 dose) in pre-filled syringe (Type-I glass), with a plunger stopper (bromobutyl) and a tip cap (polypropylene), with 2 separate needles

Pack size of 1.

- Powder (1 dose) in vial (Type-I glass), with a stopper (butyl rubber) and aluminium seal with green flip-off plastic cap + 0.5 mL solvent (1 dose) in pre-filled syringe (Type-I glass), with a plunger stopper (bromobutyl) and a tip cap (polypropylene), without needles

Pack size of 1.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

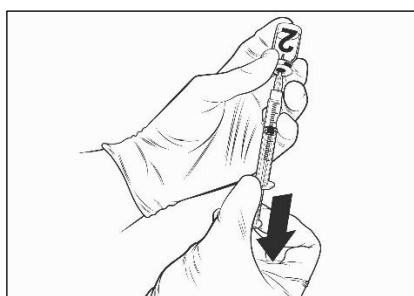
Instructions for reconstitution of the vaccine with the solvent presented in vial

Qdenga is a 2-component vaccine that consists of a vial containing lyophilised vaccine and a vial containing solvent. The lyophilised vaccine must be reconstituted with solvent prior to administration.

Use only sterile syringes for reconstitution and injection of Qdenga. Qdenga should not be mixed with other vaccines in the same syringe.

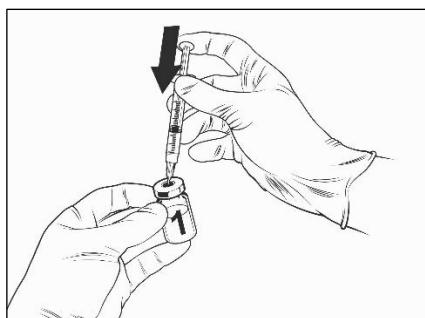
To reconstitute Qdenga, use only the solvent (0.22% sodium chloride solution) supplied with the vaccine since it is free of preservatives or other anti-viral substances. Contact with preservatives, antiseptics, detergents, and other anti-viral substances is to be avoided since they may inactivate the vaccine.

Remove the vaccine and solvent vials from the refrigerator.



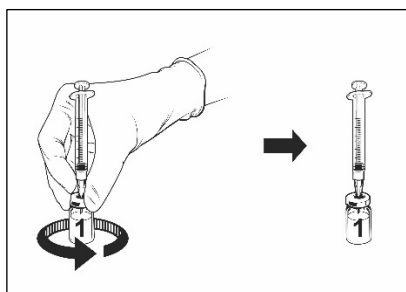
Solvent vial

- Remove the caps from both vials and clean the surface of stoppers on top of the vials using an alcohol wipe.
- Attach a sterile needle to a sterile 1 mL syringe and insert the needle into the solvent vial. The recommended needle is 23G.
- Slowly press the plunger completely down.
- Turn the vial upside down, withdraw the entire contents of the vial and continue to pull plunger out to 0.75 mL. A bubble should be seen inside of the syringe.
- Invert the syringe to bring the bubble back to the plunger.



Lyophilised vaccine vial

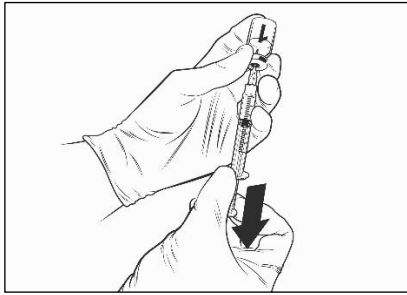
- Insert the needle of the syringe assembly into the lyophilised vaccine vial.
- Direct the flow of the solvent toward the side of the vial while slowly depressing the plunger to reduce the chance of forming bubbles.



Reconstituted vaccine

- Release your finger from the plunger and, holding the assembly on a flat surface, gently swirl the vial in both directions with the needle syringe assembly attached.
- DO NOT SHAKE. Foam and bubbles may form in the reconstituted product.
- Let the vial and syringe assembly sit for a while until the solution becomes clear. This takes about 30-60 seconds.

Following reconstitution, the resulting solution should be clear, colourless to pale yellow, and essentially free of foreign particulates. Discard the vaccine if particulates are present and/or if it appears discoloured.



Reconstituted vaccine

- Withdraw the entire volume of the reconstituted Qdenga solution with the same syringe until an air bubble appears in the syringe.
- Remove the needle syringe assembly from the vial.
- Hold the syringe with the needle pointing upwards, tap the side of the syringe to bring the air bubble to the top, discard the attached needle and replace with a new sterile needle, expel the air bubble until a small drop of the liquid forms at the top of the needle. The recommended needle is 25G 16 mm.
- Qdenga is ready to be administered by subcutaneous injection.

Qdenga should be administered immediately after reconstitution. Chemical and physical in-use stability have been demonstrated for 2 hours at room temperature (up to 32.5°C) from the time of reconstitution of the vaccine vial. After this time period, the vaccine must be discarded. Do not return it to the refrigerator. From a microbiological point of view Qdenga should be used immediately. If not used immediately, in-use storage times and conditions are the responsibility of the user.

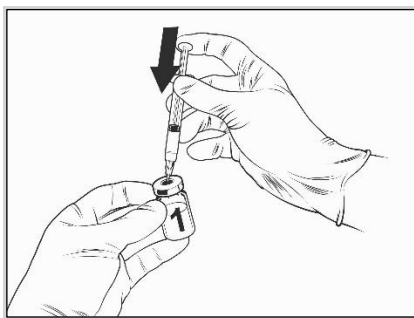
Instructions for reconstitution of the vaccine with solvent presented in pre-filled syringe

Qdenga is a 2-component vaccine that consists of a vial containing lyophilised vaccine and solvent provided in the pre-filled syringe. The lyophilised vaccine must be reconstituted with solvent prior to administration.

Qdenga should not be mixed with other vaccines in the same syringe.

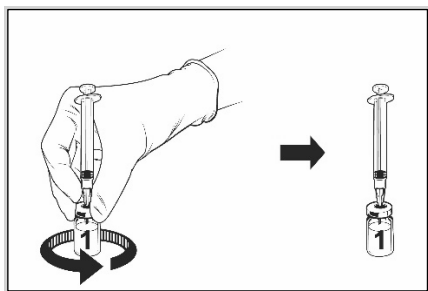
To reconstitute Qdenga, use only the solvent (0.22% sodium chloride solution) in the pre-filled syringe supplied with the vaccine since it is free of preservatives or other anti-viral substances. Contact with preservatives, antiseptics, detergents, and other anti-viral substances is to be avoided since they may inactivate the vaccine.

Remove the vaccine vial and pre-filled syringe solvent from the refrigerator.



Lyophilised vaccine vial

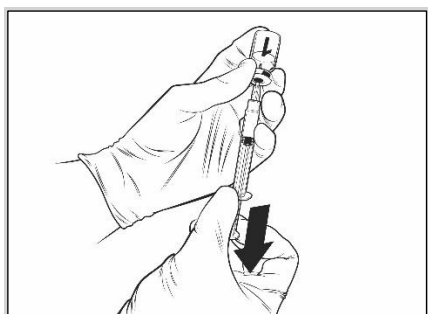
- Remove the cap from the vaccine vial and clean the surface of stopper on top of the vial using an alcohol wipe.
- Attach a sterile needle to the pre-filled syringe and insert the needle into the vaccine vial. The recommended needle is 23G.
- Direct the flow of the solvent toward the side of the vial while slowly depressing the plunger to reduce the chance of forming bubbles.



Reconstituted vaccine

- Release your finger from the plunger and, holding the assembly on a flat surface, gently swirl the vial in both directions with the needle syringe assembly attached.
- DO NOT SHAKE. Foam and bubbles may form in the reconstituted product.
- Let the vial and syringe assembly sit for a while until the solution becomes clear. This takes about 30-60 seconds.

Following reconstitution, the resulting solution should be clear, colourless to pale yellow, and essentially free of foreign particulates. Discard the vaccine if particulates are present and/or if it appears discoloured.



Reconstituted vaccine

- Withdraw the entire volume of the reconstituted Qdenga solution with the same syringe until an air bubble appears in the syringe.
- Remove the needle syringe assembly from the vial. Hold the syringe with the needle pointing upwards, tap the side of the syringe to bring the air bubble to the top, discard the attached needle and replace with a new sterile needle, expel the air bubble until a small drop of the liquid forms at the top of the needle. The recommended needle is 25G 16 mm.
- Qdenga is ready to be administered by subcutaneous injection.

Qdenga should be administered immediately after reconstitution. Chemical and physical in-use stability have been demonstrated for 2 hours at room temperature (up to 32.5°C) from the time of reconstitution of the vaccine vial. After this time period, the vaccine must be discarded. Do not return it to the refrigerator. From a microbiological point of view Qdenga should be used immediately. If not used immediately, in-use storage times and conditions are the responsibility of the user.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MARKETING AUTHORISATION HOLDER

Takeda Pharmaceuticals International AG Ireland Branch
Block 2 Miesian Plaza
50-58 Baggot Street Lower
Dublin 2
D02 HW68
Ireland

8. MARKETING AUTHORISATION NUMBER(S)

EU/1/22/1699/001
EU/1/22/1699/002
EU/1/22/1699/003
EU/1/22/1699/005

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorisation: 05 December 2022

10. DATE OF REVISION OF THE TEXT

Detailed information on this medicinal product is available on the website of the European Medicines Agency <https://www.ema.europa.eu>.

ANNEX II

- A. MANUFACTURER(S) OF THE BIOLOGICAL ACTIVE SUBSTANCE(S) AND MANUFACTURER(S) RESPONSIBLE FOR BATCH RELEASE**
- B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE**
- C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION**
- D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT**

A. MANUFACTURER(S) OF THE BIOLOGICAL ACTIVE SUBSTANCE(S) AND MANUFACTURER(S) RESPONSIBLE FOR BATCH RELEASE

Name and address of the manufacturer(s) of the biological active substance(s)

IDT Biologika GmbH
Am Pharmapark
06861 Dessau-Rosslau
Germany

Name and address of the manufacturer(s) responsible for batch release

Takeda GmbH
Production site Singen
Robert-Bosch-Str. 8
78224 Singen
Germany

B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE

Medicinal product subject to medical prescription.

- **Official batch release**

In accordance with Article 114 of Directive 2001/83/EC, the official batch release will be undertaken by a state laboratory or a laboratory designated for that purpose.

C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION

- **Periodic safety update reports (PSURs)**

The requirements for submission of PSURs for this medicinal product are set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and any subsequent updates published on the European medicines web-portal.

The marketing authorisation holder (MAH) shall submit the first PSUR for this product within 6 months following authorisation.

D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT

- **Risk management plan (RMP)**

The marketing authorisation holder (MAH) shall perform the required pharmacovigilance activities and interventions detailed in the agreed RMP presented in Module 1.8.2 of the marketing authorisation and any agreed subsequent updates of the RMP.

An updated RMP should be submitted:

- At the request of the European Medicines Agency;
- Whenever the risk management system is modified, especially as the result of new information being received that may lead to a significant change to the benefit/risk profile or as the result of an important (pharmacovigilance or risk minimisation) milestone being reached.

ANNEX III
LABELLING AND PACKAGE LEAFLET

A. LABELLING

PARTICULARS TO APPEAR ON THE OUTER PACKAGING**Powder (1 dose) in vial + solvent in vial****Pack size of 1 or 10****1. NAME OF THE MEDICINAL PRODUCT**

Qdenga powder and solvent for solution for injection
Dengue tetravalent vaccine (live, attenuated)

2. STATEMENT OF ACTIVE SUBSTANCE(S)

After reconstitution, one dose (0.5 mL) contains:

Dengue virus serotype 1 (live, attenuated): $\geq 3.3 \log_{10}$ Plaque-forming units (PFU)/dose

Dengue virus serotype 2 (live, attenuated): $\geq 2.7 \log_{10}$ PFU/dose

Dengue virus serotype 3 (live, attenuated): $\geq 4.0 \log_{10}$ PFU/dose

Dengue virus serotype 4 (live, attenuated): $\geq 4.5 \log_{10}$ PFU/dose

3. LIST OF EXCIPIENTS

Excipients:

Powder: α,α -Trehalose dihydrate, Poloxamer 407, human serum albumin, potassium dihydrogen phosphate, disodium hydrogen phosphate, potassium chloride, sodium chloride

Solvent: Sodium chloride, water for injections

4. PHARMACEUTICAL FORM AND CONTENTS

Powder and solvent for solution for injection

1 vial: powder

1 vial: solvent

1 dose (0.5 mL)

10 vials: powder

10 vials: solvent

10 x 1 dose (0.5 mL)

5. METHOD AND ROUTE(S) OF ADMINISTRATION

Subcutaneous use after reconstitution.

Read the package leaflet before use.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN
--

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING(S), IF NECESSARY
--

8. EXPIRY DATE

EXP {MM/YYYY}

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator.
Do not freeze. Store in the original package.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE
--

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Takeda Pharmaceuticals International AG Ireland Branch
Block 2 Miesian Plaza
50-58 Baggot Street Lower
Dublin 2
D02 HW68
Ireland

12. MARKETING AUTHORISATION NUMBER(S)
--

EU/1/22/1699/001
EU/1/22/1699/002

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY
--

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Justification for not including Braille accepted.

17. UNIQUE IDENTIFIER – 2D BARCODE

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA
--

PC
SN
NN

PARTICULARS TO APPEAR ON THE OUTER PACKAGING**Powder (1 dose) in vial + solvent in pre-filled syringe****Powder (1 dose) in vial + solvent in pre-filled syringe with 2 separate needles****Pack size of 1****1. NAME OF THE MEDICINAL PRODUCT**

Qdenga powder and solvent for solution for injection in pre-filled syringe
Dengue tetravalent vaccine (live, attenuated)

2. STATEMENT OF ACTIVE SUBSTANCE(S)

After reconstitution, one dose (0.5 mL) contains:

Dengue virus serotype 1 (live, attenuated): $\geq 3.3 \log_{10}$ Plaque-forming units (PFU)/dose

Dengue virus serotype 2 (live, attenuated): $\geq 2.7 \log_{10}$ PFU/dose

Dengue virus serotype 3 (live, attenuated): $\geq 4.0 \log_{10}$ PFU/dose

Dengue virus serotype 4 (live, attenuated): $\geq 4.5 \log_{10}$ PFU/dose

3. LIST OF EXCIPIENTS

Excipients:

Powder: α,α -Trehalose dihydrate, Poloxamer 407, human serum albumin, potassium dihydrogen phosphate, disodium hydrogen phosphate, potassium chloride, sodium chloride

Solvent: Sodium chloride, water for injections

4. PHARMACEUTICAL FORM AND CONTENTS

Powder and solvent for solution for injection in a pre-filled syringe

1 vial: powder

1 pre-filled syringe: solvent

1 dose (0.5 mL)

1 vial: powder

1 pre-filled syringe: solvent

2 needles

1 dose (0.5 mL)

5. METHOD AND ROUTE(S) OF ADMINISTRATION

Subcutaneous use after reconstitution.

Read the package leaflet before use.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING(S), IF NECESSARY

8. EXPIRY DATE

EXP {MM/YYYY}

9. SPECIAL STORAGE CONDITIONS

Store in a refrigerator.
Do not freeze. Store in the original package.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Takeda Pharmaceuticals International AG Ireland Branch
Block 2 Miesian Plaza
50-58 Baggot Street Lower
Dublin 2
D02 HW68
Ireland

12. MARKETING AUTHORISATION NUMBER(S)

EU/1/22/1699/003
EU/1/22/1699/005

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Justification for not including Braille accepted.

17. UNIQUE IDENTIFIER – 2D BARCODE
--

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PC
SN
NN

MINIMUM PARTICULARS TO APPEAR ON SMALL IMMEDIATE PACKAGING UNITS

Powder (1 dose) in vial

1. NAME OF THE MEDICINAL PRODUCT AND ROUTE(S) OF ADMINISTRATION
--

Qdenga
Powder for injection
Dengue tetravalent vaccine
SC

2. METHOD OF ADMINISTRATION

3. EXPIRY DATE

EXP {MM/YYYY}

4. BATCH NUMBER

Lot

5. CONTENTS BY WEIGHT, BY VOLUME OR BY UNIT
--

1 Dose

6. OTHER

MINIMUM PARTICULARS TO APPEAR ON SMALL IMMEDIATE PACKAGING UNITS

Solvent in a vial

Solvent in a pre-filled syringe

1. NAME OF THE MEDICINAL PRODUCT AND ROUTE(S) OF ADMINISTRATION
--

Solvent for Qdenga

NaCl (0.22%)

2. METHOD OF ADMINISTRATION

3. EXPIRY DATE

EXP {MM/YYYY}

4. BATCH NUMBER

Lot

5. CONTENTS BY WEIGHT, BY VOLUME OR BY UNIT
--

0.5 mL

6. OTHER

B. PACKAGE LEAFLET

Package leaflet: Information for the user

Qdenga powder and solvent for solution for injection

Dengue tetravalent vaccine (live, attenuated)

▼ This medicine is subject to additional monitoring. This will allow quick identification of new safety information. You can help by reporting any side effects you may get. See the end of section 4 for how to report side effects.

Read all of this leaflet carefully before you or your child is vaccinated because it contains important information for you.

- Keep this leaflet. You may need to read it again.
- If you have any further questions, ask your doctor, pharmacist or nurse.
- This medicine has been prescribed for you or your child only. Do not pass it on to others.
- If you or your child get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. See section 4.

What is in this leaflet

1. What Qdenga is and what it is used for
2. What you need to know before you or your child receive Qdenga
3. How Qdenga is given
4. Possible side effects
5. How to store Qdenga
6. Contents of the pack and other information

1. What Qdenga is and what it is used for

Qdenga is a vaccine. It is used to help protect you or your child against dengue. Dengue is a disease caused by dengue virus serotypes 1, 2, 3 and 4. Qdenga contains weakened versions of these 4 dengue virus serotypes so it cannot cause dengue disease.

Qdenga is given to adults, young people and children (from 4 years of age).

Qdenga should be used according to official recommendations.

How the vaccine works

Qdenga stimulates the body's natural defences (immune system). This helps to protect against the viruses that cause dengue if the body is exposed to these viruses in the future.

What dengue is

Dengue is caused by a virus.

- The virus is spread by mosquitos (Aedes mosquitos).
- If a mosquito bites someone with dengue it can pass the virus on to the next people it bites.

Dengue is not passed directly from person to person.

Signs of dengue include fever, headache, pain behind the eyes, muscle and joint pain, feeling or being sick (nausea and vomiting), swollen glands or skin rash. Signs of dengue usually last for 2 to 7 days. You can also be infected with dengue virus but show no signs of illness.

Occasionally dengue can be severe enough for you or your child to have to go to hospital and in rare cases it can cause death. Severe dengue can give you a high fever and any of the following: severe abdominal (belly) pain, persistent sickness (vomiting), rapid breathing, severe bleeding, bleeding in

the stomach, bleeding gums, feeling tired, feeling restless, coma, having fits (seizures) and organ failure.

2. What you need to know before you or your child receive Qdenga

To make sure that Qdenga is suitable for you or your child, it is important to tell your doctor, pharmacist or nurse if any of the points below apply to you or your child. If there is anything you do not understand, ask your doctor, pharmacist or nurse to explain.

Do not use Qdenga if you or your child

- are allergic to the active substances or any of the other ingredients of Qdenga (listed in section 6).
- had an allergic reaction after receiving Qdenga before. Signs of an allergic reaction may include an itchy rash, shortness of breath and swelling of the face and tongue.
- have a weak immune system (the body's natural defences). This may be due to a genetic defect or HIV infection.
- are taking a medicine that affects the immune system (such as high-dose corticosteroids or chemotherapy). Your doctor will not use Qdenga until some time after you stop treatment with this medicine.
- are pregnant or breast-feeding.

Do not use Qdenga if any of the above applies.

Warnings and precautions

Tell your doctor, pharmacist or nurse before receiving Qdenga if you or your child:

- have an infection with fever. It might be necessary to postpone the vaccination until recovery.
- have ever had any health problems when given a vaccine. Your doctor will carefully consider the risks and benefits of vaccination.
- have ever fainted from an injection. Dizziness, fainting, and sometimes falling, can happen (mostly in young people) following, or even before, any injection with a needle.

Important information about the protection provided

As with any vaccine, Qdenga may not protect everybody who receives it and protection might decrease over time. You may still get dengue fever from mosquito bites, including severe dengue illness. You must continue to protect yourself or your child against mosquito bites even after vaccination with Qdenga.

After vaccination, you should consult a doctor if you or your child believe you might have a dengue infection, and develop any of the following symptoms: high fever, severe abdominal pain, persistent vomiting, rapid breathing, bleeding gums, tiredness, restlessness and blood in vomit.

Additional protection precautions

You should take precautions to prevent mosquito bites. This includes using insect repellents, wearing protective clothing, and using mosquito nets.

Younger children

Children less than 4 years of age must not receive Qdenga.

Other medicines and Qdenga

Qdenga can be given with a hepatitis A vaccine, yellow fever vaccine or human papillomavirus vaccine at a separate injection site (another part of your body, usually the other arm) during the same visit.

Tell your doctor or pharmacist if you or your child are using, have recently used, or might use any other vaccines or medicines.

In particular, tell your doctor or pharmacist if you or your child are taking any of the following:

- Medicines that affect your body's natural defences (immune system) such as high-dose corticosteroids or chemotherapy. In this case, your doctor will not use Qdenga until some time after you stop treatment. This is because Qdenga might not work as well.
- Medicines called "immunoglobulins" or blood products containing immunoglobulins, such as blood or plasma. In this case, your doctor will not use Qdenga until 6 weeks, and preferably not for 3 months after you stop treatment. This is because Qdenga might not work as well.

Pregnancy and breast-feeding

Do not use Qdenga if you or your daughter are pregnant or breast-feeding. If you or your daughter:

- are of child-bearing age, you must take necessary precautions to avoid pregnancy for one month after Qdenga vaccination.
- think you or your daughter may be pregnant or are planning to have a baby, ask your doctor, pharmacist or nurse for advice before using Qdenga.

Driving and using machines

Qdenga has a minor influence on the ability to drive and use machines in the first days following vaccination.

Qdenga contains sodium and potassium

Qdenga contains less than 1 mmol sodium (23 mg) per 0.5 mL dose, i.e. essentially 'sodium-free'.

Qdenga contains less than 1 mmol potassium (39 mg) per 0.5 mL dose, i.e. essentially 'potassium-free'.

3. How Qdenga is given

Qdenga is given by your doctor or nurse as an injection under the skin (subcutaneous injection) in the upper arm. It must not be injected into a blood vessel.

You or your child will receive 2 injections.

The second injection is given 3 months after the first injection.

There are no data in adults above 60 years of age. Ask your doctor for advice whether it is beneficial for you to receive Qdenga.

Qdenga should be used according to official recommendations.

Instructions for preparing the vaccine intended for medical and healthcare professionals are included at the end of the leaflet.

If you or your child miss an injection of Qdenga

- If you or your child miss a scheduled injection, your doctor will decide when to give the missed injection. It is important that you or your child follow the instructions of your doctor, pharmacist or nurse about the follow-up injection.
- If you forget or are not able to go back at the scheduled time, ask your doctor, pharmacist or nurse for advice.

If you have any further questions on the use of this vaccine, ask your doctor, pharmacist or nurse.

4. Possible side effects

Like all medicines, Qdenga can cause side effects, although not everybody gets them.

Severe allergic (anaphylactic) reaction

If any of these symptoms occur after leaving the place where you or your child received an injection, **contact a doctor immediately:**

- difficulty breathing
- blueness of the tongue or lips

- a rash
- swelling of the face or throat
- low blood pressure causing dizziness or fainting
- sudden and serious feeling of illness or unease with drop in blood pressure causing dizziness and loss of consciousness, rapid heartbeat linked with breathing difficulty.

These signs or symptoms (anaphylactic reactions) usually develop soon after the injection is given and while you or your child are still in the clinic or doctor's surgery. They can also happen very rarely after receiving any vaccine.

The following side effects occurred during studies in children, young people and adults.

Very common (may affect more than 1 in 10 people):

- injection site pain
- headache
- muscle pain
- injection site redness
- generally feeling unwell
- weakness
- infections of the nose or throat
- fever

Common (may affect up to 1 in 10 people):

- injection site swelling
- pain or inflammation of the nose or throat
- injection site bruising
- injection site itching
- inflammation of throat and tonsils
- joint pain
- flu like illness

Uncommon (may affect up to 1 in 100 people):

- diarrhoea
- feeling sick
- stomach pain
- being sick (vomiting)
- injection site bleeding
- feeling lightheaded
- itchy skin
- skin rash, including blotchy or itchy skin eruptions
- hives
- tiredness
- skin colour changes at the injection site
- inflammation of the airways
- runny nose

Rare (may affect up to 1 in 1,000 people):

- small red or purple spots under your skin (petechiae)

Very rare (may affect up to 1 in 10,000 people):

- rapid swelling under the skin in areas such as the face, throat, arms and legs
- low levels of blood platelets (thrombocytopenia)

Not known (cannot be estimated from the available data):

- sudden, severe allergic (anaphylactic) reaction, with breathing difficulty, swelling,

- lightheadedness, fast heartbeat, sweating and loss of consciousness
- eye pain

Additional side effects in children 4 to 5 years of age:

Very common (may affect more than 1 in 10 people):

- decreased appetite
- feeling sleepy
- irritability

Reporting of side effects

If you get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. You can also report side effects directly via [the national reporting system](#) listed in [Appendix V](#). By reporting side effects, you can help provide more information on the safety of this medicine.

5. How to store Qdenga

Keep Qdenga out of the sight and reach of children.

Do not use Qdenga after the expiry date which is stated on the carton after EXP. The expiry date refers to the last day of that month.

Store in a refrigerator (2°C to 8°C). Do not freeze.

Keep the vaccine in the outer carton.

After mixing (reconstitution) with the solvent provided, Qdenga should be used immediately. If not used immediately, Qdenga must be used within 2 hours.

Do not throw away any medicines via wastewater or household waste. Ask your pharmacist how to throw away medicines you no longer use. These measures will help to protect the environment.

6. Contents of the pack and other information

What Qdenga contains

- After reconstitution, one dose (0.5 mL) contains:
 - Dengue virus serotype 1 (live, attenuated)*: $\geq 3.3 \log_{10}$ PFU**/dose
 - Dengue virus serotype 2 (live, attenuated)#: $\geq 2.7 \log_{10}$ PFU**/dose
 - Dengue virus serotype 3 (live, attenuated)*: $\geq 4.0 \log_{10}$ PFU**/dose
 - Dengue virus serotype 4 (live, attenuated)*: $\geq 4.5 \log_{10}$ PFU**/dose

*Produced in Vero cells by recombinant DNA technology. Genes of serotype-specific surface proteins engineered into dengue type 2 backbone. This product contains genetically modified organisms (GMOs).

#Produced in Vero cells by recombinant DNA technology.

**PFU = Plaque-forming units

- The other ingredients are: α,α -Trehalose dihydrate, Poloxamer 407, human serum albumin, potassium dihydrogen phosphate, disodium hydrogen phosphate, potassium chloride, sodium chloride, water for injections.

What Qdenga looks like and contents of the pack

Qdenga is a powder and solvent for solution for injection. Qdenga is provided as a powder in a single-dose vial and a solvent in a single-dose vial.

The powder and the solvent must be mixed together before use.

Qdenga powder and solvent for solution for injection is available in packs of 1 or 10.

Not all pack sizes might be marketed.

The powder is a white to off-white coloured compact cake.

The solvent (0.22% sodium chloride solution) is a clear, colourless liquid.

After reconstitution, Qdenga is a clear, colourless to pale yellow solution, essentially free of foreign particulates.

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Other sources of information

Detailed information on this medicine is available on the European Medicines Agency web site:
<https://www.ema.europa.eu>.

The following information is intended for healthcare professionals only:

- As with all injectable vaccines, appropriate medical treatment and supervision must always be readily available in the event of an anaphylactic reaction following the administration of Qdenga.
- Qdenga must not be mixed with other medicinal products or vaccines in the same syringe.
- Qdenga must not be administered by intravascular injection under any circumstances.
- Immunisation should be carried out by subcutaneous injection preferably in the upper arm in the region of the deltoid. Qdenga should not be administered by intramuscular injection.
- Syncope (fainting) can occur following, or even before, any vaccination as a psychogenic response to injection with a needle. Procedures should be in place to prevent injury from falling and to manage syncopal reactions.

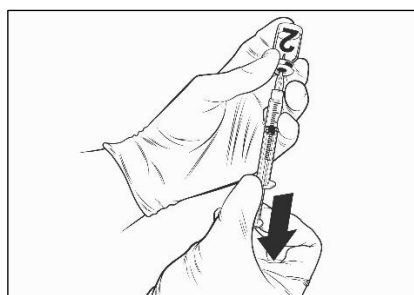
Instructions for reconstitution of the vaccine with the solvent presented in vial:

Qdenga is a 2-component vaccine that consists of a vial containing lyophilised vaccine and a vial containing solvent. The lyophilised vaccine must be reconstituted with solvent prior to administration.

Use only sterile syringes for reconstitution and injection of Qdenga. Qdenga should not be mixed with other vaccines in the same syringe.

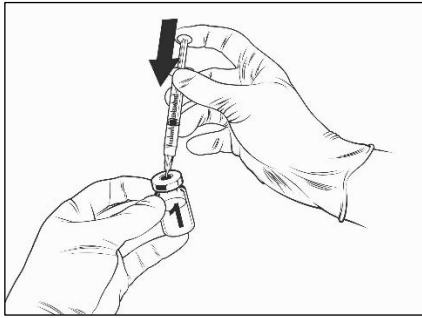
To reconstitute Qdenga, use only the solvent (0.22% sodium chloride solution) supplied with the vaccine since it is free of preservatives or other anti-viral substances. Contact with preservatives, antiseptics, detergents, and other anti-viral substances is to be avoided since they may inactivate the vaccine.

Remove the vaccine and solvent vials from the refrigerator.



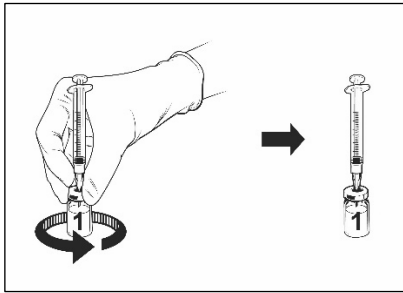
Solvent vial

- Remove the caps from both vials and clean the surface of stoppers on top of the vials using an alcohol wipe.
- Attach a sterile needle to a sterile 1 mL syringe and insert the needle into the solvent vial. The recommended needle is 23G.
- Slowly press the plunger completely down.
- Turn the vial upside down, withdraw the entire contents of the vial and continue to pull plunger out to 0.75 mL. A bubble should be seen inside of the syringe.
- Invert the syringe to bring the bubble back to the plunger.



Lyophilised vaccine vial

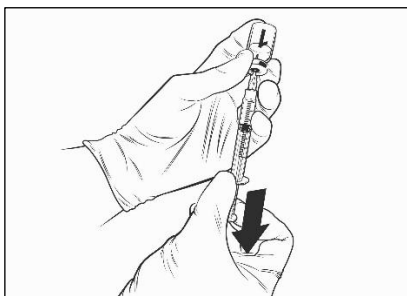
- Insert the needle of the syringe assembly into the lyophilised vaccine vial.
- Direct the flow of the solvent toward the side of the vial while slowly depressing the plunger to reduce the chance of forming bubbles.



Reconstituted vaccine

- Release your finger from the plunger and, holding the assembly on a flat surface, gently swirl the vial in both directions with the needle syringe assembly attached.
- DO NOT SHAKE. Foam and bubbles may form in the reconstituted product.
- Let the vial and syringe assembly sit for a while until the solution becomes clear. This takes about 30-60 seconds.

Following reconstitution, the resulting solution should be clear, colourless to pale yellow, and essentially free of foreign particulates. Discard the vaccine if particulates are present and/or if it appears discoloured.



Reconstituted vaccine

- Withdraw the entire volume of the reconstituted Qdenga solution with the same syringe until an air bubble appears in the syringe.
- Remove the needle syringe assembly from the vial.
- Hold the syringe with the needle pointing upwards, tap the side of the syringe to bring the air bubble to the top, discard the attached needle and replace with a new sterile needle, expel the air bubble until a small drop of the liquid forms at the top of the needle. The recommended needle is 25G 16 mm.
- Qdenga is ready to be administered by subcutaneous injection.

Qdenga should be administered immediately after reconstitution. Chemical and physical in-use stability have been demonstrated for 2 hours at room temperature (up to 32.5°C) from the time of reconstitution of the vaccine vial. After this time period, the vaccine must be discarded. Do not return it to the refrigerator. From a microbiological point of view Qdenga should be used immediately. If not used immediately, in-use storage times and conditions are the responsibility of the user.

Any unused product or waste material should be disposed of in accordance with local regulations.

Package leaflet: Information for the user

Qdenga powder and solvent for solution for injection in pre-filled syringe

Dengue tetravalent vaccine (live, attenuated)

▼ This medicine is subject to additional monitoring. This will allow quick identification of new safety information. You can help by reporting any side effects you may get. See the end of section 4 for how to report side effects.

Read all of this leaflet carefully before you or your child is vaccinated because it contains important information for you.

- Keep this leaflet. You may need to read it again.
- If you have any further questions, ask your doctor, pharmacist or nurse.
- This medicine has been prescribed for you or your child only. Do not pass it on to others.
- If you or your child get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. See section 4.

What is in this leaflet

1. What Qdenga is and what it is used for
2. What you need to know before you or your child receive Qdenga
3. How Qdenga is given
4. Possible side effects
5. How to store Qdenga
6. Contents of the pack and other information

1. What Qdenga is and what it is used for

Qdenga is a vaccine. It is used to help protect you or your child against dengue. Dengue is a disease caused by dengue virus serotypes 1, 2, 3 and 4. Qdenga contains weakened versions of these 4 dengue virus serotypes so it cannot cause dengue disease.

Qdenga is given to adults, young people and children (from 4 years of age).

Qdenga should be used according to official recommendations.

How the vaccine works

Qdenga stimulates the body's natural defences (immune system). This helps to protect against the viruses that cause dengue if the body is exposed to these viruses in the future.

What dengue is

Dengue is caused by a virus.

- The virus is spread by mosquitos (Aedes mosquitos).
- If a mosquito bites someone with dengue it can pass the virus on to the next people it bites.

Dengue is not passed directly from person to person.

Signs of dengue include fever, headache, pain behind the eyes, muscle and joint pain, feeling or being sick (nausea and vomiting), swollen glands or skin rash. Signs of dengue usually last for 2 to 7 days. You can also be infected with dengue virus but show no signs of illness.

Occasionally dengue can be severe enough for you or your child to have to go to hospital and in rare cases it can cause death. Severe dengue can give you a high fever and any of the following: severe abdominal (belly) pain, persistent sickness (vomiting), rapid breathing, severe bleeding, bleeding in

the stomach, bleeding gums, feeling tired, feeling restless, coma, having fits (seizures) and organ failure.

2. What you need to know before you or your child receive Qdenga

To make sure that Qdenga is suitable for you or your child, it is important to tell your doctor, pharmacist or nurse if any of the points below apply to you or your child. If there is anything you do not understand, ask your doctor, pharmacist or nurse to explain.

Do not use Qdenga if you or your child

- are allergic to the active substances or any of the other ingredients of Qdenga (listed in section 6).
- had an allergic reaction after receiving Qdenga before. Signs of an allergic reaction may include an itchy rash, shortness of breath and swelling of the face and tongue.
- have a weak immune system (the body's natural defences). This may be due to a genetic defect or HIV infection.
- are taking a medicine that affects the immune system (such as high-dose corticosteroids or chemotherapy). Your doctor will not use Qdenga until some time after you stop treatment with this medicine.
- are pregnant or breast-feeding.

Do not use Qdenga if any of the above applies.

Warnings and precautions

Tell your doctor, pharmacist or nurse before receiving Qdenga if you or your child:

- have an infection with fever. It might be necessary to postpone the vaccination until recovery.
- have ever had any health problems when given a vaccine. Your doctor will carefully consider the risks and benefits of vaccination.
- have ever fainted from an injection. Dizziness, fainting, and sometimes falling, can happen (mostly in young people) following, or even before, any injection with a needle.

Important information about the protection provided

As with any vaccine, Qdenga may not protect everybody who receives it and protection might decrease over time. You may still get dengue fever from mosquito bites, including severe dengue illness. You must continue to protect yourself or your child against mosquito bites even after vaccination with Qdenga.

After vaccination, you should consult a doctor if you or your child believe you might have a dengue infection, and develop any of the following symptoms: high fever, severe abdominal pain, persistent vomiting, rapid breathing, bleeding gums, tiredness, restlessness and blood in vomit.

Additional protection precautions

You should take precautions to prevent mosquito bites. This includes using insect repellents, wearing protective clothing, and using mosquito nets.

Younger children

Children less than 4 years of age must not receive Qdenga.

Other medicines and Qdenga

Qdenga can be given with a hepatitis A vaccine, yellow fever vaccine or human papillomavirus vaccine at a separate injection site (another part of your body, usually the other arm) during the same visit.

Tell your doctor or pharmacist if you or your child are using, have recently used, or might use any other vaccines or medicines.

In particular, tell your doctor or pharmacist if you or your child are taking any of the following:

- Medicines that affect your body's natural defences (immune system) such as high-dose corticosteroids or chemotherapy. In this case, your doctor will not use Qdenga until some time after you stop treatment. This is because Qdenga might not work as well.
- Medicines called "immunoglobulins" or blood products containing immunoglobulins, such as blood or plasma. In this case, your doctor will not use Qdenga until 6 weeks, and preferably not for 3 months after you stop treatment. This is because Qdenga might not work as well.

Pregnancy and breast-feeding

Do not use Qdenga if you or your daughter are pregnant or breast-feeding. If you or your daughter:

- are of child-bearing age, you must take necessary precautions to avoid pregnancy for one month after Qdenga vaccination.
- think you or your daughter may be pregnant or are planning to have a baby, ask your doctor, pharmacist or nurse for advice before using Qdenga.

Driving and using machines

Qdenga has a minor influence on the ability to drive and use machines in the first days following vaccination.

Qdenga contains sodium and potassium

Qdenga contains less than 1 mmol sodium (23 mg) per 0.5 mL dose, i.e. essentially 'sodium-free'.

Qdenga contains less than 1 mmol potassium (39 mg) per 0.5 mL dose, i.e. essentially 'potassium-free'.

3. How Qdenga is given

Qdenga is given by your doctor or nurse as an injection under the skin (subcutaneous injection) in the upper arm. It must not be injected into a blood vessel.

You or your child will receive 2 injections.

The second injection is given 3 months after the first injection.

There are no data in adults above 60 years of age. Ask your doctor for advice whether it is beneficial for you to receive Qdenga.

Qdenga should be used according to official recommendations.

Instructions for preparing the vaccine intended for medical and healthcare professionals are included at the end of the leaflet.

If you or your child miss an injection of Qdenga

- If you or your child miss a scheduled injection, your doctor will decide when to give the missed injection. It is important that you or your child follow the instructions of your doctor, pharmacist or nurse about the follow-up injection.
- If you forget or are not able to go back at the scheduled time, ask your doctor, pharmacist or nurse for advice.

If you have any further questions on the use of this vaccine, ask your doctor, pharmacist or nurse.

4. Possible side effects

Like all medicines, Qdenga can cause side effects, although not everybody gets them.

Severe allergic (anaphylactic) reaction

If any of these symptoms occur after leaving the place where you or your child received an injection, **contact a doctor immediately:**

- difficulty breathing
- blueness of the tongue or lips

- a rash
- swelling of the face or throat
- low blood pressure causing dizziness or fainting
- sudden and serious feeling of illness or unease with drop in blood pressure causing dizziness and loss of consciousness, rapid heartbeat linked with breathing difficulty.

These signs or symptoms (anaphylactic reactions) usually develop soon after the injection is given and while you or your child are still in the clinic or doctor's surgery. They can also happen very rarely after receiving any vaccine.

The following side effects occurred during studies in children, young people and adults.

Very common (may affect more than 1 in 10 people):

- injection site pain
- headache
- muscle pain
- injection site redness
- generally feeling unwell
- weakness
- infections of the nose or throat
- fever

Common (may affect up to 1 in 10 people):

- injection site swelling
- pain or inflammation of the nose or throat
- injection site bruising
- injection site itching
- inflammation of throat and tonsils
- joint pain
- flu like illness

Uncommon (may affect up to 1 in 100 people):

- diarrhoea
- feeling sick
- stomach pain
- being sick (vomiting)
- injection site bleeding
- feeling lightheaded
- itchy skin
- skin rash, including blotchy or itchy skin eruptions
- hives
- tiredness
- skin colour changes at the injection site
- inflammation of the airways
- runny nose

Rare (may affect up to 1 in 1,000 people):

- small red or purple spots under your skin (petechiae)

Very rare (may affect up to 1 in 10,000 people):

- rapid swelling under the skin in areas such as the face, throat, arms and legs
- low levels of blood platelets (thrombocytopenia)

Not known (cannot be estimated from the available data):

- sudden, severe allergic (anaphylactic) reaction, with breathing difficulty, swelling,

- lightheadedness, fast heartbeat, sweating and loss of consciousness
- eye pain

Additional side effects in children 4 to 5 years of age:

Very common (may affect more than 1 in 10 people):

- decreased appetite
- feeling sleepy
- irritability

Reporting of side effects

If you get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. You can also report side effects directly via [the national reporting system](#) listed in [Appendix V](#). By reporting side effects, you can help provide more information on the safety of this medicine.

5. How to store Qdenga

Keep Qdenga out of the sight and reach of children.

Do not use Qdenga after the expiry date which is stated on the carton after EXP. The expiry date refers to the last day of that month.

Store in a refrigerator (2°C to 8°C). Do not freeze.

Keep the vaccine in the outer carton.

After mixing (reconstitution) with the solvent provided, Qdenga should be used immediately. If not used immediately, Qdenga must be used within 2 hours.

Do not throw away any medicines via wastewater or household waste. Ask your pharmacist how to throw away medicines you no longer use. These measures will help to protect the environment.

6. Contents of the pack and other information

What Qdenga contains

- After reconstitution, one dose (0.5 mL) contains:
 - Dengue virus serotype 1 (live, attenuated)*: $\geq 3.3 \log_{10}$ PFU**/dose
 - Dengue virus serotype 2 (live, attenuated)#: $\geq 2.7 \log_{10}$ PFU**/dose
 - Dengue virus serotype 3 (live, attenuated)*: $\geq 4.0 \log_{10}$ PFU**/dose
 - Dengue virus serotype 4 (live, attenuated)*: $\geq 4.5 \log_{10}$ PFU**/dose

*Produced in Vero cells by recombinant DNA technology. Genes of serotype-specific surface proteins engineered into dengue type 2 backbone. This product contains genetically modified organisms (GMOs).

#Produced in Vero cells by recombinant DNA technology.

**PFU = Plaque-forming units

- The other ingredients are: α,α -Trehalose dihydrate, Poloxamer 407, human serum albumin, potassium dihydrogen phosphate, disodium hydrogen phosphate, potassium chloride, sodium chloride, water for injections.

What Qdenga looks like and contents of the pack

Qdenga is a powder and solvent for solution for injection. Qdenga is provided as a powder in a single-dose vial and a solvent in pre-filled syringe with 2 separate needles or with no needle. The powder and the solvent must be mixed together before use.

Qdenga powder and solvent for solution for injection in pre-filled syringe is available in packs of 1.

Not all pack sizes might be marketed.

The powder is a white to off-white coloured compact cake.

The solvent (0.22% sodium chloride solution) is a clear, colourless liquid.

After reconstitution, Qdenga is a clear, colourless to pale yellow solution, essentially free of foreign particulates.

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Other sources of information

Detailed information on this medicine is available on the European Medicines Agency web site:
<https://www.ema.europa.eu>.

The following information is intended for healthcare professionals only:

- As with all injectable vaccines, appropriate medical treatment and supervision must always be readily available in the event of an anaphylactic reaction following the administration of Qdenga.
- Qdenga must not be mixed with other medicinal products or vaccines in the same syringe.
- Qdenga must not be administered by intravascular injection under any circumstances.
- Immunisation should be carried out by subcutaneous injection preferably in the upper arm in the region of the deltoid. Qdenga should not be administered by intramuscular injection.
- Syncope (fainting) can occur following, or even before, any vaccination as a psychogenic response to injection with a needle. Procedures should be in place to prevent injury from falling and to manage syncopal reactions.

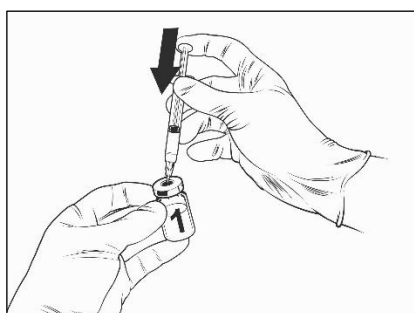
Instructions for reconstitution of the vaccine with solvent presented in pre-filled syringe:

Qdenga is a 2-component vaccine that consists of a vial containing lyophilised vaccine and solvent provided in the pre-filled syringe. The lyophilised vaccine must be reconstituted with solvent prior to administration.

Qdenga should not be mixed with other vaccines in the same syringe.

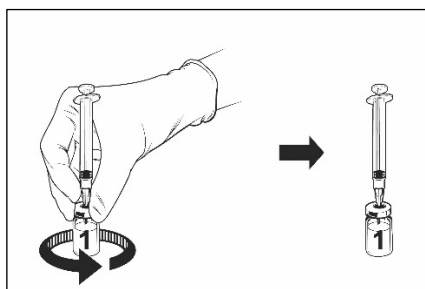
To reconstitute Qdenga, use only the solvent (0.22% sodium chloride solution) in the pre-filled syringe supplied with the vaccine since it is free of preservatives or other anti-viral substances. Contact with preservatives, antiseptics, detergents, and other anti-viral substances is to be avoided since they may inactivate the vaccine.

Remove the vaccine vial and pre-filled syringe solvent from the refrigerator.



Lyophilised vaccine vial

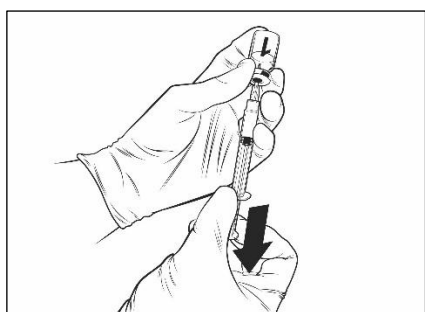
- Remove the cap from the vaccine vial and clean the surface of stopper on top of the vial using an alcohol wipe.
- Attach a sterile needle to the pre-filled syringe and insert the needle into the vaccine vial. The recommended needle is 23G.
- Direct the flow of the solvent toward the side of the vial while slowly depressing the plunger to reduce the chance of forming bubbles.



Reconstituted vaccine

- Release your finger from the plunger and, holding the assembly on a flat surface, gently swirl the vial in both directions with the needle syringe assembly attached.
- **DO NOT SHAKE.** Foam and bubbles may form in the reconstituted product.
- Let the vial and syringe assembly sit for a while until the solution becomes clear. This takes about 30-60 seconds.

Following reconstitution, the resulting solution should be clear, colourless to pale yellow, and essentially free of foreign particulates. Discard the vaccine if particulates are present and/or if it appears discoloured.



Reconstituted vaccine

- Withdraw the entire volume of the reconstituted Qdenga solution with the same syringe until an air bubble appears in the syringe.
- Remove the needle syringe assembly from the vial.
- Hold the syringe with the needle pointing upwards, tap the side of the syringe to bring the air bubble to the top, discard the attached needle and replace with a new sterile needle, expel the air bubble until a small drop of the liquid forms at the top of the needle. The recommended needle is 25G 16 mm.
- Qdenga is ready to be administered by subcutaneous injection.

Qdenga should be administered immediately after reconstitution. Chemical and physical in-use stability have been demonstrated for 2 hours at room temperature (up to 32.5°C) from the time of reconstitution of the vaccine vial. After this time period, the vaccine must be discarded. Do not return it to the refrigerator. From a microbiological point of view Qdenga should be used immediately. If not used immediately, in-use storage times and conditions are the responsibility of the user.

Any unused product or waste material should be disposed of in accordance with local regulations.